Phase changes

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Account for energy transfer during a phase change without a temperature rise.

SUCCESS CRITERIA - I can...

1. Identify a constant-temperature phase-change interval.
2. Use Q=mL for that interval.
3. Separate warming energy from phase-change energy.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “phase change” and “latent heat” to explain your model or result. Sentence starters are optional.

phase change: a change between states such as solid and liquid

latent heat: energy per mass needed for a specified phase change

temperature: a measure of thermal state; it is not an energy amount

01 Warm-up: bring back an earlier idea

Two objects at 280 K have different masses and materials. Must their internal energies be equal?
Give two reasons.

Sentence starter: Both ... . They differ in ... because ... .

Marzano check - circle: ① ② ③

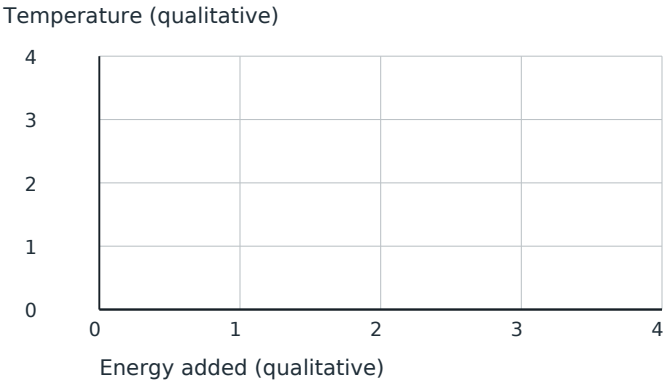
02 Worked example: explain the model

Reference: Q = mL; Q = mc ΔT.
Use mL during a phase change at constant temperature, and mc ΔT outside that interval. L is latent heat.

Worked reasoning: Melting 2 kg of a substance with latent heat 100 J/kg requires 200 J. During ideal melting at fixed pressure, added energy changes phase without raising temperature.

Your turn: Sketch temperature versus added energy through warming and melting.
Mark the plateau.

Sentence starter: The ... in my model represents ... . Its label shows ... .



Marzano check - circle: ① ② ③

Phase changes

antocciphysics

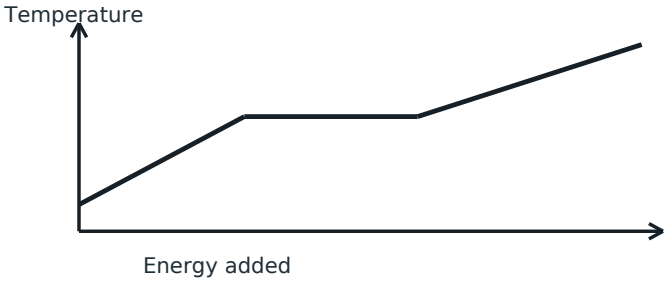
Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Read and interpret the model

On the heating curve, identify the flat interval.
Explain why added energy does not raise temperature during that interval.

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .



The horizontal axis shows energy added. The vertical axis shows temperature.

Marzano check - circle: 1 2 3

04 Apply the physics

A beaker contains 0.10 kg of ice at 0°C. First melt it, then warm the water to 20°C.
Use L_f = 334,000 J/kg and c_{water} = 4200 J/(kg·°C).
Label the two stages, calculate each energy transfer and find the total.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Stage	Equation	Energy (J)
Melt at 0°C	____	____
Warm 0→20°C	____	____

Marzano check - circle: 1 2 3

05 Explain your reasoning

A student uses mc ΔT for a melting interval with ΔT=0 and concludes no energy is required.
Explain the missing term.

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

With right positive, a spring has k=200 N/m and displacement x=-0.06 m.
Find signed spring force and direction.
Mark equilibrium, displacement and restoring-force direction.

Word help: displacement = change from initial position to final position, with direction

Sentence starter: I found ... [unit]. This value means ... in this situation.



Mark equilibrium and displacement x.

Marzano check - circle: 1 2 3

Phase changes

antocciphysics

Continue down each column.

Sentence starters are optional. You may plan in any language.

07 Exit check

A 0.20 kg vapor sample releases 45,000 J as it condenses to liquid at constant temperature.
Label the outgoing energy arrow.
Find its latent heat of condensation and state the sign of Q for the sample.

Sentence starter: I found … [unit]. This value means … in this situation.

Initial T / phase: _____ Final T / phase: _____

Sample

Add an energy arrow; label mass, c or L as needed.

Marzano check - circle: 1 2 3

08 STRETCH: test the model

A 0.2 kg ice sample at 0°C melts and the water warms to 10°C.
Use L_f=334000 J/kg and c_{water}=4200 J/(kg·°C).
Find total heat.
Label the energy arrow and identify what changes in the sample.

Sentence starter: I found … [unit]. This value means … in this situation.

Initial T / phase: _____ Final T / phase: _____

Sample

Add an energy arrow; label mass, c or L as needed.

Marzano check - circle: 1 2 3